

SOLUTION

1. Represent the Tic-Tac-Toe board as a list of 9 positions.
2. The human player uses **X** and the computer uses **O**.
3. Generate all possible legal moves.
4. Use the **Minimax algorithm** to evaluate possible moves.
5. Maintain two values:
 - **Alpha:** Best value currently guaranteed for the maximizing player.
 - **Beta:** Best value currently guaranteed for the minimizing player.
6. If $\alpha \geq \beta$, stop exploring the remaining branches because they cannot improve the result.
7. Select the move with the highest score for the computer.
8. Continue until there is a winner or the game ends in a draw.

Advantage: Alpha-Beta Pruning produces the same optimal result as Minimax while evaluating fewer game-tree nodes.